



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Computing

Semester 01 2025/2026

Subject : Database Programming (SECP3623)
Section : Section 1-2
Task : Project I
Due : 25th November 2025 (10%)

Name	Matric No.
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GUI KAH SIN	A23CS0080
SABRINA HENG WEI QI	A23CS0265
LAU YAN KAI	A23CS0098

Instructions

Each group must submit **one (1)** project only.

Your submission must include:

1. **Project Report in PDF format** (named as LeaderName_GroupNum_ProjectI.pdf).
2. The **demo video URL** (YouTube or Google Drive link) **inside the report**.
 - o Make sure the link is working, publicly accessible, and does not require login.
3. Include all SQL script files (.sql) by attaching them directly, compressing them into one zip folder, or providing a valid OneDrive sharing link inside the report.

Presentation Video: <https://youtu.be/gJTp1mYCpvk>

Marking Rubric

SQL Implementation (CLO1)

Criteria	Excellent (5)	Good (4)	Fair (3)	Weak (2)	Poor (1)	Marks
Backend SQL Implementation (DDL/DML/Queries)	All SQL statements execute flawlessly; covers full range – DDL, DML, SELECT, joins, subqueries, functions.	Most SQL correct; small syntax or logic issues.	Several working statements but limited variety.	Few working statements; repetitive or basic logic.	Non-functional / incomplete SQL.	
Advanced Data Logic (Filtering, Aggregation, Subquery)	Uses complex logic (nested queries, GROUP BY, HAVING, CASE, conditions) effectively.	Correct use of aggregations with minor gaps.	Some queries too simple or misapplied.	Few aggregations or logical errors.	Not attempted.	
Indexing & Optimization	Correctly creates BTREE and TEXT indexes; shows analysis before/after; interprets performance gain.	Indexes implemented; partial analysis.	Indexing attempted without analysis.	Incomplete index usage.	Not attempted.	
Transaction Control & Front-End Logic Integration	Demonstrates atomic operations (START TRANSACTION, COMMIT, ROLLBACK); explains how it fits system workflow (front-end use case e.g. booking/purchase).	Transaction logic corrects but limited integration explanation.	Partial transaction implementation.	Minimal attempt, unclear scenario.	Missing transaction logic.	
Subtotal						

Teamwork and Collaboration (CLO3)

Criteria	Excellent (5)	Good (4)	Fair (3)	Weak (2)	Poor (1)	Marks
Team Coordination & Contribution	Clear task division; balanced participation; strong collaboration evident in report and presentation; each member contributes to SQL work.	Good teamwork and participation with minor imbalance.	Some collaboration; limited task clarity.	Minimal teamwork; dominated by few members.	No teamwork evidence; incomplete group submission.	
Subtotal						

Documentation & Presentation (CLO4)

Criteria	Excellent (5)	Good (4)	Fair (3)	Weak (2)	Poor (1)	Marks
Report Documentation	Clear and complete report including intro, SQL summary, query screenshots, optimization, transaction explanation, and reflection.	Report complete with minor formatting or clarity issues.	Adequate report but lacks details or screenshots.	Weakly structured or incomplete sections.	Missing or very poor documentation.	
Slide Quality & Visual Communication	Slides are professional, organised, visually clear, and consistent with project flow.	Well-prepared slides with minor design issues.	Slides understandable but inconsistent layout.	Poorly structured or cluttered slides.	Missing or unreadable slides.	
Verbal Presentation	Excellent delivery, confident speakers, balanced time usage, fluent explanation, and effective Q&A handling.	Good delivery; some coordination issues.	Understandable but low energy; moderate Q&A.	Disorganized delivery or weak explanations.	Unable to present or answer questions.	
Subtotal						

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INTRODUCTION

1.0 Background

One of the most important tasks in managing universities and complying with its official standards is tracking students' attendance. In the past, attendance was recorded manually, people used paper check-in forms or shouted their names in the class. These outdated codes are unreliable. They are slow, error-prone, and difficult to check or audit in the future. This delay makes it difficult for universities to quickly find and support students who skip classes, and slows down all office work.

In a university with many courses at the same time and thousands of students, we need a reliable, scalable which can handle growth and automated system. An automated system is needed to accurately track students' attendance and ensure that the university follows all classroom participation standards.

The project aims to solve this problem by focussing on the basic back-end functions of the Academic Attendance Tracking System. Our main goal is to establish a centralised and efficient data repository for all attendance information. The system is not just a simple manual record storage but also designed to handle complex data processing. We are focussing on developing a relational database model, and implementing all the necessary SQL programming infrastructure. The system will be able to handle all routine activities such as capturing students' classroom check-in and providing complete attendance records.

1.1 Objectives

The main goals of this project are to build the functional database for the Academic Attendance Tracking System by demonstrating all the required SQL skills using MySQL:

No	Skills	Description
1	Database setup (DDL)	To design and create all the necessary tables, and make sure the tables have the basic rules like PRIMARY KEY, FOREIGN KEY and to check the information is correct.
2	Data management (DML)	To insert, update and delete records, and also to use transaction like COMMIT and ROLLBACK to keep the data consistent.
3	Advanced queries (DQL)	To write complex search commands that support system functions like aggregation, join, and subqueries.

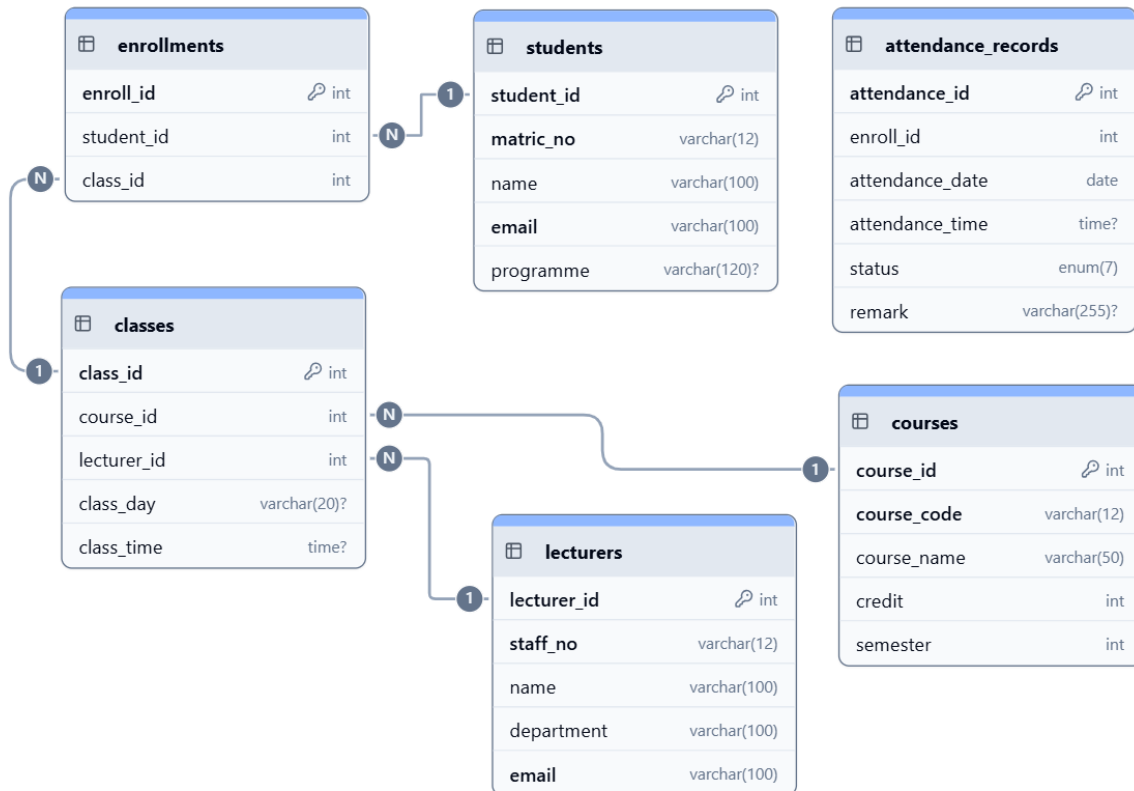
No	Skills	Description
4	Performance improvement	To create internal structure like BTREE and TEXT indexes to make searches faster, and to analyse exactly how much performance improves.
5	Reporting and presentation	To work as a team to create a detailed report, and give clear presentation.

1.2 Team Members and Assigned Roles

Team Member	Roles
Brendan Chia Yan Fei A23CS0211	▪ Advanced DQL ▪ Joins ▪ Subqueries
Gui Kah Sin A23CS0080	▪ DML implementation ▪ DML manipulation ▪ Basic DQL
Sabrina Heng Wei Qi A23CS0265	▪ Database design ▪ Database creation ▪ DDL manipulation
Lau Yan Kai A23CS0098	▪ Transaction ▪ Indexing ▪ Optimisation analysis ▪ Advanced DQL

SQL Implementation

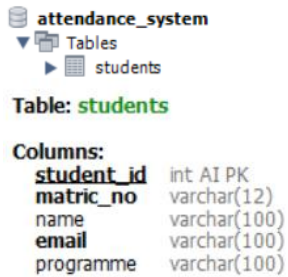
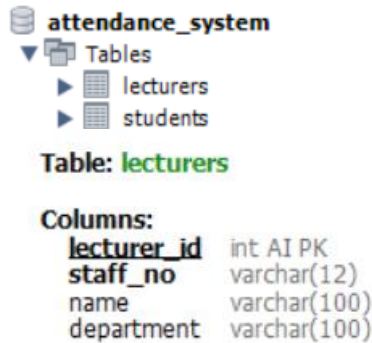
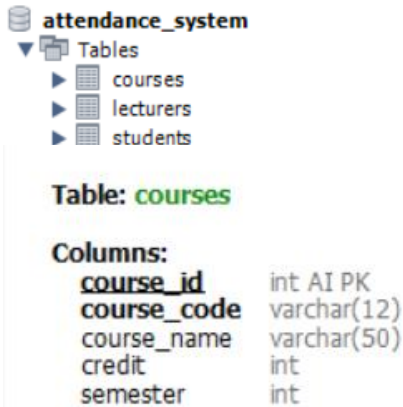
2.0 Database Design

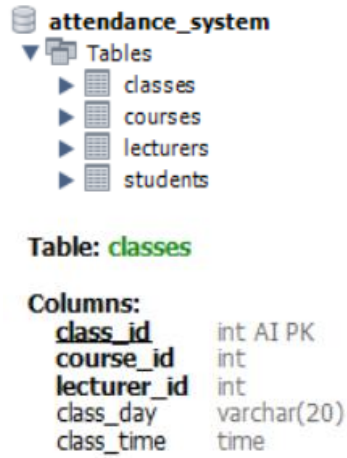
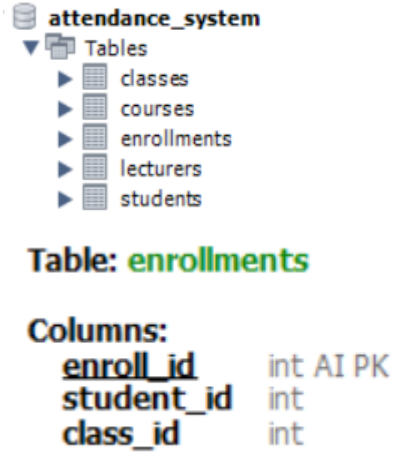
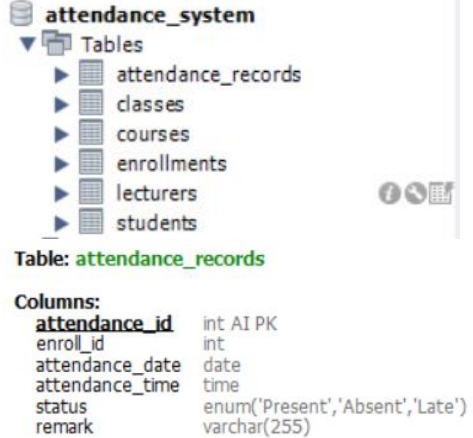


2.1 DDL Implementation

This section will show the database schema design and the SQL Data Definition Language (DDL) used to create the attendance system backend.

SQL Code	Output
1. Database Creation	
CREATE DATABASE is used to create a database that named attendance_system. By using IF NOT EXISTS, this able to check that the database has been created or not to prevent duplicates databases error.	
<pre>CREATE DATABASE IF NOT EXISTS attendance_system; USE attendance_system;</pre>	

SQL Code	Output										
2. Create students table											
Use CREATE TABLE to create a table named students and using IF NOT EXISTS to make sure that the table are not created before. PRIMARY KEY used to set the primary key for each table and AUTO_INCREMENT is used to let it auto create the primary key when next line of database exists. Others like datatype such as INT, VARCHAR are used to identify each rows type. Lastly, by using constraints NOT NULL able to prevent the table to be empty and UNIQUE will make sure that the content does not duplicate. These are all apply to creating tables code below.											
<pre>CREATE TABLE IF NOT EXISTS students (student_id INT PRIMARY KEY AUTO_INCREMENT, matric_no VARCHAR (12) NOT NULL UNIQUE, name VARCHAR(100) NOT NULL, email VARCHAR(100) NOT NULL UNIQUE, programme VARCHAR (100) NOT NULL);</pre>	 Table: students Columns: <table> <tr> <td>student_id</td> <td>int AI PK</td> </tr> <tr> <td>matric_no</td> <td>varchar(12)</td> </tr> <tr> <td>name</td> <td>varchar(100)</td> </tr> <tr> <td>email</td> <td>varchar(100)</td> </tr> <tr> <td>programme</td> <td>varchar(100)</td> </tr> </table>	student_id	int AI PK	matric_no	varchar(12)	name	varchar(100)	email	varchar(100)	programme	varchar(100)
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matric_no	varchar(12)										
name	varchar(100)										
email	varchar(100)										
programme	varchar(100)										
3. Create lecturers table											
<pre>CREATE TABLE IF NOT EXISTS lecturers (lecturer_id INT PRIMARY KEY AUTO_INCREMENT, staff_no VARCHAR(12) NOT NULL UNIQUE, name VARCHAR(100) NOT NULL, department VARCHAR(100) NOT NULL);</pre>	 Table: lecturers Columns: <table> <tr> <td>lecturer_id</td> <td>int AI PK</td> </tr> <tr> <td>staff_no</td> <td>varchar(12)</td> </tr> <tr> <td>name</td> <td>varchar(100)</td> </tr> <tr> <td>department</td> <td>varchar(100)</td> </tr> </table>	lecturer_id	int AI PK	staff_no	varchar(12)	name	varchar(100)	department	varchar(100)		
lecturer_id	int AI PK										
staff_no	varchar(12)										
name	varchar(100)										
department	varchar(100)										
4. Create courses table											
<pre>CREATE TABLE IF NOT EXISTS courses (course_id INT PRIMARY KEY AUTO_INCREMENT, course_code VARCHAR(12) NOT NULL UNIQUE, course_name VARCHAR(50) NOT NULL, credit INT NOT NULL, semester INT NOT NULL);</pre>	 Table: courses Columns: <table> <tr> <td>course_id</td> <td>int AI PK</td> </tr> <tr> <td>course_code</td> <td>varchar(12)</td> </tr> <tr> <td>course_name</td> <td>varchar(50)</td> </tr> <tr> <td>credit</td> <td>int</td> </tr> <tr> <td>semester</td> <td>int</td> </tr> </table>	course_id	int AI PK	course_code	varchar(12)	course_name	varchar(50)	credit	int	semester	int
course_id	int AI PK										
course_code	varchar(12)										
course_name	varchar(50)										
credit	int										
semester	int										

SQL Code	Output												
5. Create classes table													
<pre>CREATE TABLE IF NOT EXISTS classes (class_id INT PRIMARY KEY AUTO_INCREMENT, course_id INT NOT NULL, lecturer_id INT NOT NULL, class_day VARCHAR(20), class_time TIME, FOREIGN KEY(course_id) REFERENCES courses(course_id), FOREIGN KEY (lecturer_id) REFERENCES lecturers(lecturer_id));</pre>	 Table: classes Columns: <table> <tr> <td><u>class_id</u></td> <td>int AI PK</td> </tr> <tr> <td>course_id</td> <td>int</td> </tr> <tr> <td>lecturer_id</td> <td>int</td> </tr> <tr> <td>class_day</td> <td>varchar(20)</td> </tr> <tr> <td>class_time</td> <td>time</td> </tr> </table>	<u>class_id</u>	int AI PK	course_id	int	lecturer_id	int	class_day	varchar(20)	class_time	time		
<u>class_id</u>	int AI PK												
course_id	int												
lecturer_id	int												
class_day	varchar(20)												
class_time	time												
6. Create enrolments table													
<pre>CREATE TABLE IF NOT EXISTS enrollments (enroll_id INT PRIMARY KEY AUTO_INCREMENT, student_id INT NOT NULL, class_id INT NOT NULL, FOREIGN KEY (student_id) REFERENCES students(student_id), FOREIGN KEY (class_id) REFERENCES classes(class_id));</pre>	 Table: enrollments Columns: <table> <tr> <td><u>enroll_id</u></td> <td>int AI PK</td> </tr> <tr> <td>student_id</td> <td>int</td> </tr> <tr> <td>class_id</td> <td>int</td> </tr> </table>	<u>enroll_id</u>	int AI PK	student_id	int	class_id	int						
<u>enroll_id</u>	int AI PK												
student_id	int												
class_id	int												
7. Create attendance_records table													
<pre>CREATE TABLE IF NOT EXISTS attendance_records (attendance_id INT PRIMARY KEY AUTO_INCREMENT, enroll_id INT NOT NULL, attendance_date DATE NOT NULL, attendance_time TIME, #change it so that it can store null for absent reason status ENUM('Present','Absent','Late') NOT NULL, remark VARCHAR (255) DEFAULT NULL);</pre>	 Table: attendance_records Columns: <table> <tr> <td><u>attendance_id</u></td> <td>int AI PK</td> </tr> <tr> <td>enroll_id</td> <td>int</td> </tr> <tr> <td>attendance_date</td> <td>date</td> </tr> <tr> <td>attendance_time</td> <td>time</td> </tr> <tr> <td>status</td> <td>enum('Present','Absent','Late')</td> </tr> <tr> <td>remark</td> <td>varchar(255)</td> </tr> </table>	<u>attendance_id</u>	int AI PK	enroll_id	int	attendance_date	date	attendance_time	time	status	enum('Present','Absent','Late')	remark	varchar(255)
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enroll_id	int												
attendance_date	date												
attendance_time	time												
status	enum('Present','Absent','Late')												
remark	varchar(255)												

SQL Code	Output										
8. ALTER TABLE – Add email to lecturers table											
<pre>ALTER TABLE lecturers ADD COLUMN email VARCHAR(100) NOT NULL UNIQUE;</pre>	Table: lecturers Columns: <table> <tr><td>lecturer_id</td><td>int AI PK</td></tr> <tr><td>staff_no</td><td>varchar(12)</td></tr> <tr><td>name</td><td>varchar(100)</td></tr> <tr><td>department</td><td>varchar(100)</td></tr> <tr><td>email</td><td>varchar(100)</td></tr> </table>	lecturer_id	int AI PK	staff_no	varchar(12)	name	varchar(100)	department	varchar(100)	email	varchar(100)
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name	varchar(100)										
department	varchar(100)										
email	varchar(100)										
9. ALTER TABLE – Change programme length											
<pre>ALTER TABLE students MODIFY COLUMN programme VARCHAR(120);</pre>	Table: students Columns: <table> <tr><td>student_id</td><td>int AI PK</td></tr> <tr><td>matric_no</td><td>varchar(12)</td></tr> <tr><td>name</td><td>varchar(100)</td></tr> <tr><td>email</td><td>varchar(100)</td></tr> <tr><td>programme</td><td>varchar(120)</td></tr> </table>	student_id	int AI PK	matric_no	varchar(12)	name	varchar(100)	email	varchar(100)	programme	varchar(120)
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matric_no	varchar(12)										
name	varchar(100)										
email	varchar(100)										
programme	varchar(120)										
10. DROP TABLE – drop temporary table if exist											
<pre>#drop a temporary table DROP TABLE IF EXISTS temp_attendance;</pre>											

2.2 DML Implementation

SQL Code	Output																																																							
1. Insert 10 students' records																																																								
<pre>#insert 10 students records INSERT INTO students (student_id, matric_no, name, email, programme) VALUES (1001, 'A23CS0009', 'Aisyah', 'aisyah@graduate.utm.my', 'Computer Networks and Security'), (1003, 'A23CS0023', 'Balqis', 'balqis@graduate.utm.my', 'Software Engineering'), (1008, 'A23CS0018', 'Brendan Chia', 'brendan@graduate.utm.my', 'Bioinformatics'), (1017, 'A23CS0037', 'Fatin Afifah', 'fatin@graduate.utm.my', 'Computer Networks and Security'), (1023, 'A23CS0080', 'Gui Kah Sin', 'kahsin@graduate.utm.my', 'Data Engineering'), (1056, 'A23CS0098', 'Lau Yan Kai', 'yanikai@graduate.utm.my', 'Software Engineering'), (1080, 'A23CS0128', 'Matthew', 'matthew@graduate.utm.my', 'Data Engineering'), (1107, 'A23CS00135', 'Muhammad Iqram', 'iqram@graduate.utm.my', 'Bioinformatics'), (1129, 'A23CS0265', 'Sabrina Heng', 'sabrina@graduate.utm.my', 'Data Engineering'), (1212, 'A23CS0239', 'Yusuf', 'yusuf@graduate.utm.my', 'Graphics and Multimedia Software');</pre>	<table><tr><th>student_id</th><th>matric_no</th><th>name</th><th>email</th><th>programme</th></tr><tr><td>1001</td><td>A23CS0009</td><td>Aisyah</td><td>aisyah@graduate.utm.my</td><td>Computer Networks and Security</td></tr><tr><td>1003</td><td>A23CS0023</td><td>Balqis</td><td>balqis@graduate.utm.my</td><td>Software Engineering</td></tr><tr><td>1008</td><td>A23CS0018</td><td>Brendan Chia</td><td>brendan@graduate.utm.my</td><td>Bioinformatics</td></tr><tr><td>1017</td><td>A23CS0037</td><td>Fatin Afifah</td><td>fatin@graduate.utm.my</td><td>Computer Networks and Security</td></tr><tr><td>1023</td><td>A23CS0080</td><td>Gui Kah Sin</td><td>kahsin@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1056</td><td>A23CS0098</td><td>Lau Yan Kai</td><td>yanikai@graduate.utm.my</td><td>Software Engineering</td></tr><tr><td>1080</td><td>A23CS0128</td><td>Matthew</td><td>matthew@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1107</td><td>A23CS00135</td><td>Muhammad Iqram</td><td>iqram@graduate.utm.my</td><td>Bioinformatics</td></tr><tr><td>1129</td><td>A23CS0265</td><td>Sabrina Heng</td><td>sabrina@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1212</td><td>A23CS0239</td><td>Yusuf</td><td>yusuf@graduate.utm.my</td><td>Graphics and Multimedia Software</td></tr></table>	student_id	matric_no	name	email	programme	1001	A23CS0009	Aisyah	aisyah@graduate.utm.my	Computer Networks and Security	1003	A23CS0023	Balqis	balqis@graduate.utm.my	Software Engineering	1008	A23CS0018	Brendan Chia	brendan@graduate.utm.my	Bioinformatics	1017	A23CS0037	Fatin Afifah	fatin@graduate.utm.my	Computer Networks and Security	1023	A23CS0080	Gui Kah Sin	kahsin@graduate.utm.my	Data Engineering	1056	A23CS0098	Lau Yan Kai	yanikai@graduate.utm.my	Software Engineering	1080	A23CS0128	Matthew	matthew@graduate.utm.my	Data Engineering	1107	A23CS00135	Muhammad Iqram	iqram@graduate.utm.my	Bioinformatics	1129	A23CS0265	Sabrina Heng	sabrina@graduate.utm.my	Data Engineering	1212	A23CS0239	Yusuf	yusuf@graduate.utm.my	Graphics and Multimedia Software
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2. Insert 10 lectures records																																																								
<pre>#insert 10 lecturers records INSERT INTO lecturers (lecturer_id, staff_no, name, department, email) VALUES (3001, '10231', 'Dr. Azlan', 'Department of Computer Networks & Cybersecurity', 'azlan@utm.my'), (3002, '12124', 'Dr. Siti', 'Department of Software Engineering', 'siti@utm.my'), (3003, '9742', 'Dr. Lee', 'Department of Computational Biology & Bioinformatics', 'lee@utm.my'), (3004, '12111', 'Dr. Sean', 'Department of Artificial Intelligence & Data Science', 'seahcs@utm.my'), (3005, '15946', 'Dr. Hani', 'Department of Multimedia', 'hani@utm.my'), (3006, '7963', 'Dr. Faridah', 'Department of Multimedia', 'faridah@utm.my'), (3007, '16508', 'Dr. Lim', 'Department of Software Engineering', 'lim@utm.my'), (3008, '9832', 'Dr. Nurin', 'Department of Multimedia', 'nurin@utm.my'), (3009, '13126', 'Dr. Nies', 'Department of Computational Biology & Bioinformatics', 'nies@utm.my'), (3010, '7864', 'Dr. Najmi', 'Department of Artificial Intelligence & Data Science', 'najmi@utm.my');</pre>	<table><tr><th>lecturer_id</th><th>staff_no</th><th>name</th><th>department</th><th>email</th></tr><tr><td>3001</td><td>10231</td><td>Dr. Azlan</td><td>Department of Computer Networks & Cybersec...</td><td>azlan@utm.my</td></tr><tr><td>3002</td><td>12124</td><td>Dr. Siti</td><td>Department of Software Engineering</td><td>siti@utm.my</td></tr><tr><td>3003</td><td>9742</td><td>Dr. Lee</td><td>Department of Computational Biology & Bioinfor...</td><td>lee@utm.my</td></tr><tr><td>3004</td><td>12111</td><td>Dr. Sean</td><td>Department of Artificial Intelligence & Data Scie...</td><td>seahcs@utm.my</td></tr><tr><td>3005</td><td>15946</td><td>Dr. Hani</td><td>Department of Multimedia</td><td>hani@utm.my</td></tr><tr><td>3006</td><td>7963</td><td>Dr. Faridah</td><td>Department of Multimedia</td><td>faridah@utm.my</td></tr><tr><td>3007</td><td>16508</td><td>Dr. Lim</td><td>Department of Software Engineering</td><td>lim@utm.my</td></tr><tr><td>3008</td><td>9832</td><td>Dr. Nurin</td><td>Department of Multimedia</td><td>nurin@utm.my</td></tr><tr><td>3009</td><td>13126</td><td>Dr. Nies</td><td>Department of Computational Biology & Bioinfor...</td><td>nies@utm.my</td></tr><tr><td>3010</td><td>7864</td><td>Dr. Najmi</td><td>Department of Artificial Intelligence & Data Scie...</td><td>najmi@utm.my</td></tr></table>	lecturer_id	staff_no	name	department	email	3001	10231	Dr. Azlan	Department of Computer Networks & Cybersec...	azlan@utm.my	3002	12124	Dr. Siti	Department of Software Engineering	siti@utm.my	3003	9742	Dr. Lee	Department of Computational Biology & Bioinfor...	lee@utm.my	3004	12111	Dr. Sean	Department of Artificial Intelligence & Data Scie...	seahcs@utm.my	3005	15946	Dr. Hani	Department of Multimedia	hani@utm.my	3006	7963	Dr. Faridah	Department of Multimedia	faridah@utm.my	3007	16508	Dr. Lim	Department of Software Engineering	lim@utm.my	3008	9832	Dr. Nurin	Department of Multimedia	nurin@utm.my	3009	13126	Dr. Nies	Department of Computational Biology & Bioinfor...	nies@utm.my	3010	7864	Dr. Najmi	Department of Artificial Intelligence & Data Scie...	najmi@utm.my
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3002	12124	Dr. Siti	Department of Software Engineering	siti@utm.my																																																				
3003	9742	Dr. Lee	Department of Computational Biology & Bioinfor...	lee@utm.my																																																				
3004	12111	Dr. Sean	Department of Artificial Intelligence & Data Scie...	seahcs@utm.my																																																				
3005	15946	Dr. Hani	Department of Multimedia	hani@utm.my																																																				
3006	7963	Dr. Faridah	Department of Multimedia	faridah@utm.my																																																				
3007	16508	Dr. Lim	Department of Software Engineering	lim@utm.my																																																				
3008	9832	Dr. Nurin	Department of Multimedia	nurin@utm.my																																																				
3009	13126	Dr. Nies	Department of Computational Biology & Bioinfor...	nies@utm.my																																																				
3010	7864	Dr. Najmi	Department of Artificial Intelligence & Data Scie...	najmi@utm.my																																																				

SQL Code	Output																																																																											
3. Insert 10 courses records																																																																												
<pre>#insert 10 courses records INSERT INTO courses (course_id, course_code, course_name, credit, semester) VALUES (1, 'SECJ1013', 'Programming Language I', 3, 1), (3, 'SECJ1023', 'Discrete Structure', 3, 1), (102, 'SECJ2013', 'Database Programming', 3, 5), (24, 'SECJ2023', 'System Analysis and Design (WBL)', 3, 2), (43, 'SECJ4033', 'Human Computer Interaction', 3, 2), (56, 'SECJ3013', 'Network Communications', 3, 3), (72, 'SECJ3023', 'Software Engineering (WBL)', 4, 3), (87, 'SECJ3033', 'Operating Systems', 3, 4), (113, 'SECJ4013', 'Artificial Intelligence', 3, 5), (96, 'SECJ4023', 'Data Mining', 3, 4), (43, 'SECJ4033', 'Human Computer Interaction', 3, 2);</pre>	<table><tr><th>course_id</th><th>course_code</th><th>course_name</th><th>credit</th><th>semester</th></tr><tr><td>1</td><td>SECJ1013</td><td>Programming Language I</td><td>3</td><td>1</td></tr><tr><td>3</td><td>SECJ1023</td><td>Discrete Structure</td><td>3</td><td>1</td></tr><tr><td>24</td><td>SECJ2023</td><td>System Analysis and Design (WBL)</td><td>3</td><td>2</td></tr><tr><td>43</td><td>SECJ4033</td><td>Human Computer Interaction</td><td>3</td><td>2</td></tr><tr><td>56</td><td>SECJ3013</td><td>Network Communications</td><td>3</td><td>3</td></tr><tr><td>72</td><td>SECJ3023</td><td>Software Engineering (WBL)</td><td>4</td><td>3</td></tr><tr><td>87</td><td>SECJ3033</td><td>Operating Systems</td><td>3</td><td>4</td></tr><tr><td>96</td><td>SECJ4023</td><td>Data Mining</td><td>3</td><td>4</td></tr><tr><td>102</td><td>SECJ2013</td><td>Database Programming</td><td>3</td><td>5</td></tr><tr><td>113</td><td>SECJ4013</td><td>Artificial Intelligence</td><td>3</td><td>5</td></tr></table>	course_id	course_code	course_name	credit	semester	1	SECJ1013	Programming Language I	3	1	3	SECJ1023	Discrete Structure	3	1	24	SECJ2023	System Analysis and Design (WBL)	3	2	43	SECJ4033	Human Computer Interaction	3	2	56	SECJ3013	Network Communications	3	3	72	SECJ3023	Software Engineering (WBL)	4	3	87	SECJ3033	Operating Systems	3	4	96	SECJ4023	Data Mining	3	4	102	SECJ2013	Database Programming	3	5	113	SECJ4013	Artificial Intelligence	3	5																				
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87	SECJ3033	Operating Systems	3	4																																																																								
96	SECJ4023	Data Mining	3	4																																																																								
102	SECJ2013	Database Programming	3	5																																																																								
113	SECJ4013	Artificial Intelligence	3	5																																																																								
4. Insert 14 classes records																																																																												
<pre>#insert 14 classes records INSERT INTO classes (class_id, course_id, lecturer_id, class_day, class_time) VALUES (1, 1, 3001, 'Monday', '09:00:00'), (2, 3, 3002, 'Tuesday', '10:00:00'), (3, 102, 3003, 'Wednesday', '14:00:00'), (4, 24, 3004, 'Thursday', '11:00:00'), (5, 56, 3005, 'Friday', '15:00:00'), (6, 72, 3006, 'Monday', '08:00:00'), (7, 87, 3007, 'Tuesday', '09:00:00'), (8, 113, 3008, 'Wednesday', '10:00:00'), (9, 96, 3009, 'Thursday', '08:00:00'), (10, 43, 3010, 'Friday', '15:00:00'), (11, 1, 3001, 'Wednesday', '11:00:00'), (12, 96, 3009, 'Monday', '09:00:00'), (13, 24, 3004, 'Wednesday', '11:00:00'), (14, 113, 3008, 'Tuesday', '08:00:00');</pre>	<table><tr><th>class_id</th><th>course_id</th><th>lecturer_id</th><th>class_day</th><th>class_time</th></tr><tr><td>1</td><td>1</td><td>3001</td><td>Monday</td><td>09:00:00</td></tr><tr><td>2</td><td>3</td><td>3002</td><td>Tuesday</td><td>10:00:00</td></tr><tr><td>3</td><td>102</td><td>3003</td><td>Wednesday</td><td>14:00:00</td></tr><tr><td>4</td><td>24</td><td>3004</td><td>Thursday</td><td>11:00:00</td></tr><tr><td>5</td><td>56</td><td>3005</td><td>Friday</td><td>15:00:00</td></tr><tr><td>6</td><td>72</td><td>3006</td><td>Monday</td><td>08:00:00</td></tr><tr><td>7</td><td>87</td><td>3007</td><td>Tuesday</td><td>09:00:00</td></tr><tr><td>8</td><td>113</td><td>3008</td><td>Wednesday</td><td>10:00:00</td></tr><tr><td>9</td><td>96</td><td>3009</td><td>Thursday</td><td>08:00:00</td></tr><tr><td>10</td><td>43</td><td>3010</td><td>Friday</td><td>15:00:00</td></tr><tr><td>11</td><td>1</td><td>3001</td><td>Wednesday</td><td>11:00:00</td></tr><tr><td>12</td><td>96</td><td>3009</td><td>Monday</td><td>09:00:00</td></tr><tr><td>13</td><td>24</td><td>3004</td><td>Wednesday</td><td>11:00:00</td></tr><tr><td>14</td><td>113</td><td>3008</td><td>Tuesday</td><td>08:00:00</td></tr></table>	class_id	course_id	lecturer_id	class_day	class_time	1	1	3001	Monday	09:00:00	2	3	3002	Tuesday	10:00:00	3	102	3003	Wednesday	14:00:00	4	24	3004	Thursday	11:00:00	5	56	3005	Friday	15:00:00	6	72	3006	Monday	08:00:00	7	87	3007	Tuesday	09:00:00	8	113	3008	Wednesday	10:00:00	9	96	3009	Thursday	08:00:00	10	43	3010	Friday	15:00:00	11	1	3001	Wednesday	11:00:00	12	96	3009	Monday	09:00:00	13	24	3004	Wednesday	11:00:00	14	113	3008	Tuesday	08:00:00
class_id	course_id	lecturer_id	class_day	class_time																																																																								
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13	24	3004	Wednesday	11:00:00																																																																								
14	113	3008	Tuesday	08:00:00																																																																								
5. Insert 2 enrollments records per students																																																																												
<pre>#insert 2 enrollments records per student INSERT INTO enrollments (enroll_id, student_id, class_id) VALUES (1, 1001, 1), (2, 1001, 3), (3, 1003, 2), (4, 1003, 4), (5, 1008, 3), (6, 1008, 5), (7, 1017, 1), (8, 1017, 2), (9, 1023, 5), (10, 1023, 6), (11, 1056, 7), (12, 1056, 8), (13, 1080, 6), (14, 1080, 9), (15, 1107, 7), (16, 1107, 10), (17, 1129, 4), (18, 1129, 8), (19, 1212, 9), (20, 1212, 10);</pre>	<table><tr><th>enroll_id</th><th>student_id</th><th>class_id</th></tr><tr><td>1</td><td>1001</td><td>1</td></tr><tr><td>2</td><td>1001</td><td>3</td></tr><tr><td>5</td><td>1008</td><td>3</td></tr><tr><td>6</td><td>1008</td><td>5</td></tr><tr><td>7</td><td>1017</td><td>1</td></tr><tr><td>8</td><td>1017</td><td>2</td></tr><tr><td>9</td><td>1023</td><td>5</td></tr><tr><td>10</td><td>1023</td><td>6</td></tr><tr><td>11</td><td>1056</td><td>7</td></tr><tr><td>12</td><td>1056</td><td>8</td></tr><tr><td>13</td><td>1080</td><td>6</td></tr><tr><td>14</td><td>1080</td><td>9</td></tr><tr><td>16</td><td>1107</td><td>10</td></tr><tr><td>17</td><td>1129</td><td>4</td></tr><tr><td>18</td><td>1129</td><td>8</td></tr><tr><td>19</td><td>1212</td><td>9</td></tr><tr><td>20</td><td>1212</td><td>10</td></tr></table>	enroll_id	student_id	class_id	1	1001	1	2	1001	3	5	1008	3	6	1008	5	7	1017	1	8	1017	2	9	1023	5	10	1023	6	11	1056	7	12	1056	8	13	1080	6	14	1080	9	16	1107	10	17	1129	4	18	1129	8	19	1212	9	20	1212	10																					
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SQL Code	Output																																																																																																																														
6. Insert attendance records for 2 days per students																																																																																																																															
<pre>#Insert attendance records for 2 days for students INSERT INTO attendance_records (attendance_id, enroll_id, attendance_date, attendance_time, status, remark) VALUES (1, 1, '2025-11-10', '09:00:00', 'Present', NULL), (2, 2, '2025-11-11', '09:10:00', 'Late', 'Arrived 10 mins late'), (3, 3, '2025-11-10', '14:00:00', 'Present', 'Sick'), (4, 4, '2025-11-11', '14:00:00', 'Present', NULL), (5, 5, '2025-11-10', '10:00:00', 'Present', NULL), (6, 6, '2025-11-11', '10:00:00', 'Present', NULL), (7, 7, '2025-11-10', '11:05:00', 'Late', 'Arrived 5 mins late'), (8, 8, '2025-11-11', '11:00:00', 'Present', NULL), (9, 9, '2025-11-10', '14:00:00', 'Present', NULL), (10, 10, '2025-11-11', '14:00:00', 'Present', NULL), (11, 11, '2025-11-10', NULL, 'Absent', 'Family reason'), (12, 12, '2025-11-11', '15:00:00', 'Present', NULL), (13, 13, '2025-11-10', '08:00:00', 'Present', NULL), (14, 14, '2025-11-11', '08:15:00', 'Late', 'Traffic delay'), (15, 15, '2025-11-10', '09:00:00', 'Present', NULL), (16, 16, '2025-11-11', '09:03:00', 'Present', NULL), (17, 17, '2025-11-10', '10:00:00', 'Present', NULL), (18, 18, '2025-11-11', '10:00:00', 'Present', NULL), (19, 19, '2025-11-10', '13:10:00', 'Present', NULL), (20, 20, '2025-11-11', '13:00:00', 'Present', NULL);</pre>	<table><tr><th>attendance_id</th><th>enroll_id</th><th>attendance_date</th><th>attendance_time</th><th>status</th><th>remark</th></tr><tr><td>1</td><td>1</td><td>2025-11-10</td><td>09:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>2</td><td>2</td><td>2025-11-11</td><td>09:10:00</td><td>Late</td><td>Arrived 10 mins late</td></tr><tr><td>3</td><td>3</td><td>2025-11-10</td><td>14:00:00</td><td>Present</td><td>Sick</td></tr><tr><td>4</td><td>4</td><td>2025-11-11</td><td>14:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>5</td><td>5</td><td>2025-11-10</td><td>10:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>6</td><td>6</td><td>2025-11-11</td><td>10:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>7</td><td>7</td><td>2025-11-10</td><td>11:05:00</td><td>Late</td><td>Arrived 5 mins late</td></tr><tr><td>8</td><td>8</td><td>2025-11-11</td><td>11:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>9</td><td>9</td><td>2025-11-10</td><td>14:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>10</td><td>10</td><td>2025-11-11</td><td>14:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>11</td><td>11</td><td>2025-11-10</td><td>NULL</td><td>Absent</td><td>Family reason</td></tr><tr><td>12</td><td>12</td><td>2025-11-11</td><td>15:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>13</td><td>13</td><td>2025-11-10</td><td>08:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>14</td><td>14</td><td>2025-11-11</td><td>08:15:00</td><td>Late</td><td>Traffic delay</td></tr><tr><td>15</td><td>15</td><td>2025-11-10</td><td>09:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>16</td><td>16</td><td>2025-11-11</td><td>09:03:00</td><td>Present</td><td>NULL</td></tr><tr><td>17</td><td>17</td><td>2025-11-10</td><td>10:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>18</td><td>18</td><td>2025-11-11</td><td>10:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>19</td><td>19</td><td>2025-11-10</td><td>13:10:00</td><td>Present</td><td>NULL</td></tr><tr><td>20</td><td>20</td><td>2025-11-11</td><td>13:00:00</td><td>Present</td><td>NULL</td></tr></table>	attendance_id	enroll_id	attendance_date	attendance_time	status	remark	1	1	2025-11-10	09:00:00	Present	NULL	2	2	2025-11-11	09:10:00	Late	Arrived 10 mins late	3	3	2025-11-10	14:00:00	Present	Sick	4	4	2025-11-11	14:00:00	Present	NULL	5	5	2025-11-10	10:00:00	Present	NULL	6	6	2025-11-11	10:00:00	Present	NULL	7	7	2025-11-10	11:05:00	Late	Arrived 5 mins late	8	8	2025-11-11	11:00:00	Present	NULL	9	9	2025-11-10	14:00:00	Present	NULL	10	10	2025-11-11	14:00:00	Present	NULL	11	11	2025-11-10	NULL	Absent	Family reason	12	12	2025-11-11	15:00:00	Present	NULL	13	13	2025-11-10	08:00:00	Present	NULL	14	14	2025-11-11	08:15:00	Late	Traffic delay	15	15	2025-11-10	09:00:00	Present	NULL	16	16	2025-11-11	09:03:00	Present	NULL	17	17	2025-11-10	10:00:00	Present	NULL	18	18	2025-11-11	10:00:00	Present	NULL	19	19	2025-11-10	13:10:00	Present	NULL	20	20	2025-11-11	13:00:00	Present	NULL
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20	20	2025-11-11	13:00:00	Present	NULL																																																																																																																										
7. UPDATE – update <i>status</i> and <i>remark</i> for those who arrived late based on the minute of <i>attendance_time</i> that more than '00'																																																																																																																															
<pre># Update attendance status if student arrived late but show present UPDATE attendance_records SET status = 'Late' WHERE attendance_time IS NOT NULL AND MINUTE(attendance_time) > 0; # Add automated remark for late students if no remark exists or the status just updated UPDATE attendance_records SET remark = CONCAT('Arrived ', MINUTE(attendance_time), ' mins late') WHERE status = 'Late' AND (remark IS NULL OR remark = '');</pre>	<table><tr><th>attendance_id</th><th>enroll_id</th><th>attendance_date</th><th>attendance_time</th><th>status</th><th>remark</th></tr><tr><td>1</td><td>1</td><td>2025-11-10</td><td>09:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>2</td><td>2</td><td>2025-11-11</td><td>09:10:00</td><td>Late</td><td>Arrived 10 mins late</td></tr><tr><td>3</td><td>3</td><td>2025-11-10</td><td>14:00:00</td><td>Present</td><td>Sick</td></tr><tr><td>4</td><td>4</td><td>2025-11-11</td><td>14:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>5</td><td>5</td><td>2025-11-10</td><td>10:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>6</td><td>6</td><td>2025-11-11</td><td>10:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>7</td><td>7</td><td>2025-11-10</td><td>11:05:00</td><td>Late</td><td>Arrived 5 mins late</td></tr><tr><td>8</td><td>8</td><td>2025-11-11</td><td>11:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>9</td><td>9</td><td>2025-11-10</td><td>14:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>10</td><td>10</td><td>2025-11-11</td><td>14:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>11</td><td>11</td><td>2025-11-10</td><td>NULL</td><td>Absent</td><td>Family reason</td></tr><tr><td>12</td><td>12</td><td>2025-11-11</td><td>15:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>13</td><td>13</td><td>2025-11-10</td><td>08:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>14</td><td>14</td><td>2025-11-11</td><td>08:15:00</td><td>Late</td><td>Traffic delay</td></tr><tr><td>15</td><td>15</td><td>2025-11-10</td><td>09:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>16</td><td>16</td><td>2025-11-11</td><td>09:03:00</td><td>Late</td><td>Arrived 3 mins late</td></tr><tr><td>17</td><td>17</td><td>2025-11-10</td><td>10:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>18</td><td>18</td><td>2025-11-11</td><td>10:00:00</td><td>Present</td><td>NULL</td></tr><tr><td>19</td><td>19</td><td>2025-11-10</td><td>13:10:00</td><td>Late</td><td>Arrived 10 mins late</td></tr><tr><td>20</td><td>20</td><td>2025-11-11</td><td>13:00:00</td><td>Present</td><td>NULL</td></tr></table>	attendance_id	enroll_id	attendance_date	attendance_time	status	remark	1	1	2025-11-10	09:00:00	Present	NULL	2	2	2025-11-11	09:10:00	Late	Arrived 10 mins late	3	3	2025-11-10	14:00:00	Present	Sick	4	4	2025-11-11	14:00:00	Present	NULL	5	5	2025-11-10	10:00:00	Present	NULL	6	6	2025-11-11	10:00:00	Present	NULL	7	7	2025-11-10	11:05:00	Late	Arrived 5 mins late	8	8	2025-11-11	11:00:00	Present	NULL	9	9	2025-11-10	14:00:00	Present	NULL	10	10	2025-11-11	14:00:00	Present	NULL	11	11	2025-11-10	NULL	Absent	Family reason	12	12	2025-11-11	15:00:00	Present	NULL	13	13	2025-11-10	08:00:00	Present	NULL	14	14	2025-11-11	08:15:00	Late	Traffic delay	15	15	2025-11-10	09:00:00	Present	NULL	16	16	2025-11-11	09:03:00	Late	Arrived 3 mins late	17	17	2025-11-10	10:00:00	Present	NULL	18	18	2025-11-11	10:00:00	Present	NULL	19	19	2025-11-10	13:10:00	Late	Arrived 10 mins late	20	20	2025-11-11	13:00:00	Present	NULL
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8. DELETE – To remove students who have withdrawn from the program, we must first drop the foreign key constraint in the <i>enrollments</i> table since it references <i>students</i> . Then, we re-add the constraint with ON DELETE CASCADE so that when a student is deleted, all their related enrolment records are automatically removed. This prevents foreign key conflicts and ensures data integrity. Next, delete an enrolment record for the student who drops the class.																																																																																																																															
<pre># The current foreign key doesn't allow deleting a student because enrollments reference them. # To apply ON DELETE CASCADE, we first need to drop the existing foreign key. ALTER TABLE enrollments DROP FOREIGN KEY enrollments_ibfk_1; # then recreate it with ON DELETE CASCADE so that deleting a student will also removes their enrollments automatically. ALTER TABLE enrollments ADD CONSTRAINT enrollments_ibfk_1 FOREIGN KEY (student_id) REFERENCES students(student_id) ON DELETE CASCADE; # Now can delete a student record that have withdrawn from the programme DELETE FROM students WHERE student_id = 1001; # Delete an enrollment if a student dropped a class DELETE FROM enrollments where enroll_id = 15;</pre>	<table><tr><th>student_id</th><th>matric_no</th><th>name</th><th>email</th><th>programme</th></tr><tr><td>1001</td><td>A23CS0009</td><td>Aisyah</td><td>aisyah@graduate.utm.my</td><td>Computer Networks and Security</td></tr><tr><td>1008</td><td>A23CS0018</td><td>Brendan Chia</td><td>brendan@graduate.utm.my</td><td>Bioinformatics</td></tr><tr><td>1017</td><td>A23CS0037</td><td>Fatin Affah</td><td>fatin@graduate.utm.my</td><td>Computer Networks and Security</td></tr><tr><td>1023</td><td>A23CS0080</td><td>Gul Kah Sin</td><td>kahsin@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1056</td><td>A23CS0098</td><td>Lau Yan Kai</td><td>yankai@graduate.utm.my</td><td>Software Engineering</td></tr><tr><td>1080</td><td>A23CS0128</td><td>Matthew</td><td>matthew@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1107</td><td>A23CS00135</td><td>Muhammad Iqram</td><td>iqram@graduate.utm.my</td><td>Bioinformatics</td></tr><tr><td>1129</td><td>A23CS0265</td><td>Sabrina Heng</td><td>sabrina@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1212</td><td>A23CS0239</td><td>Yusuf</td><td>yusuf@graduate.utm.my</td><td>Graphics and Multimedia Software</td></tr></table> <table><tr><th>enroll_id</th><th>student_id</th><th>class_id</th></tr><tr><td>1</td><td>1001</td><td>1</td></tr><tr><td>2</td><td>1001</td><td>3</td></tr><tr><td>5</td><td>1008</td><td>3</td></tr><tr><td>6</td><td>1008</td><td>5</td></tr><tr><td>7</td><td>1017</td><td>1</td></tr><tr><td>8</td><td>1017</td><td>2</td></tr><tr><td>9</td><td>1023</td><td>5</td></tr><tr><td>10</td><td>1023</td><td>6</td></tr><tr><td>11</td><td>1056</td><td>7</td></tr><tr><td>12</td><td>1056</td><td>8</td></tr><tr><td>13</td><td>1080</td><td>6</td></tr><tr><td>14</td><td>1080</td><td>9</td></tr><tr><td>16</td><td>1107</td><td>10</td></tr><tr><td>17</td><td>1129</td><td>4</td></tr><tr><td>18</td><td>1129</td><td>8</td></tr><tr><td>19</td><td>1212</td><td>9</td></tr><tr><td>20</td><td>1212</td><td>10</td></tr></table>	student_id	matric_no	name	email	programme	1001	A23CS0009	Aisyah	aisyah@graduate.utm.my	Computer Networks and Security	1008	A23CS0018	Brendan Chia	brendan@graduate.utm.my	Bioinformatics	1017	A23CS0037	Fatin Affah	fatin@graduate.utm.my	Computer Networks and Security	1023	A23CS0080	Gul Kah Sin	kahsin@graduate.utm.my	Data Engineering	1056	A23CS0098	Lau Yan Kai	yankai@graduate.utm.my	Software Engineering	1080	A23CS0128	Matthew	matthew@graduate.utm.my	Data Engineering	1107	A23CS00135	Muhammad Iqram	iqram@graduate.utm.my	Bioinformatics	1129	A23CS0265	Sabrina Heng	sabrina@graduate.utm.my	Data Engineering	1212	A23CS0239	Yusuf	yusuf@graduate.utm.my	Graphics and Multimedia Software	enroll_id	student_id	class_id	1	1001	1	2	1001	3	5	1008	3	6	1008	5	7	1017	1	8	1017	2	9	1023	5	10	1023	6	11	1056	7	12	1056	8	13	1080	6	14	1080	9	16	1107	10	17	1129	4	18	1129	8	19	1212	9	20	1212	10																						
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2.3 Transaction

SQL Code	Output																																																							
1. Register a new student																																																								
<pre>START TRANSACTION; INSERT INTO students (matric_no, name, email, programme) VALUES ('A23CS9000', 'Transaction Test Student', 'test@graduate.utm.my', 'Software Engineering');</pre>																																																								
2. Immediately enrol the student into class ID 1																																																								
<pre>INSERT INTO enrollments (student_id, class_id) VALUES (LAST_INSERT_ID(), 1); COMMIT;</pre>	<table><tr><th>student_id</th><th>name</th><th>matric_no</th><th>class_id</th><th>enroll_id</th></tr><tr><td>1213</td><td>Transaction Test Student</td><td>A23CS9000</td><td>1</td><td>21</td></tr></table>	student_id	name	matric_no	class_id	enroll_id	1213	Transaction Test Student	A23CS9000	1	21																																													
student_id	name	matric_no	class_id	enroll_id																																																				
1213	Transaction Test Student	A23CS9000	1	21																																																				
3. Accidentally delete a student																																																								
<pre>START TRANSACTION; DELETE FROM students WHERE student_id = 1212; SELECT * FROM students;</pre>	<table><tr><th>student_id</th><th>matric_no</th><th>name</th><th>email</th><th>programme</th></tr><tr><td>1001</td><td>A23CS0009</td><td>Aisyah</td><td>aisyah@graduate.utm.my</td><td>Computer Networks and Security</td></tr><tr><td>1008</td><td>A23CS0018</td><td>Brendan Chia</td><td>brendan@graduate.utm.my</td><td>Bioinformatics</td></tr><tr><td>1017</td><td>A23CS0037</td><td>Fatin Alfah</td><td>fatin@graduate.utm.my</td><td>Computer Networks and Security</td></tr><tr><td>1023</td><td>A23CS0080</td><td>Gui Kah Sin</td><td>kahsin@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1056</td><td>A23CS0098</td><td>Lau Yan Kai</td><td>yanikai@graduate.utm.my</td><td>Software Engineering</td></tr><tr><td>1080</td><td>A23CS0128</td><td>Matthew</td><td>matthew@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1107</td><td>A23CS0135</td><td>Muhammad Iqram</td><td>iqram@graduate.utm.my</td><td>Bioinformatics</td></tr><tr><td>1129</td><td>A23CS0265</td><td>Sabrina Heng</td><td>sabrina@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1213</td><td>A23CS9000</td><td>Transaction Test Student</td><td>test@graduate.utm.my</td><td>Software Engineering</td></tr></table>	student_id	matric_no	name	email	programme	1001	A23CS0009	Aisyah	aisyah@graduate.utm.my	Computer Networks and Security	1008	A23CS0018	Brendan Chia	brendan@graduate.utm.my	Bioinformatics	1017	A23CS0037	Fatin Alfah	fatin@graduate.utm.my	Computer Networks and Security	1023	A23CS0080	Gui Kah Sin	kahsin@graduate.utm.my	Data Engineering	1056	A23CS0098	Lau Yan Kai	yanikai@graduate.utm.my	Software Engineering	1080	A23CS0128	Matthew	matthew@graduate.utm.my	Data Engineering	1107	A23CS0135	Muhammad Iqram	iqram@graduate.utm.my	Bioinformatics	1129	A23CS0265	Sabrina Heng	sabrina@graduate.utm.my	Data Engineering	1213	A23CS9000	Transaction Test Student	test@graduate.utm.my	Software Engineering					
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4. Rollback to the previous action																																																								
<pre>ROLLBACK; SELECT * FROM students;</pre>	<table><tr><th>student_id</th><th>matric_no</th><th>name</th><th>email</th><th>programme</th></tr><tr><td>1001</td><td>A23CS0009</td><td>Aisyah</td><td>aisyah@graduate.utm.my</td><td>Computer Networks and Security</td></tr><tr><td>1008</td><td>A23CS0018</td><td>Brendan Chia</td><td>brendan@graduate.utm.my</td><td>Bioinformatics</td></tr><tr><td>1017</td><td>A23CS0037</td><td>Fatin Alfah</td><td>fatin@graduate.utm.my</td><td>Computer Networks and Security</td></tr><tr><td>1023</td><td>A23CS0080</td><td>Gui Kah Sin</td><td>kahsin@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1056</td><td>A23CS0098</td><td>Lau Yan Kai</td><td>yanikai@graduate.utm.my</td><td>Software Engineering</td></tr><tr><td>1080</td><td>A23CS0128</td><td>Matthew</td><td>matthew@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1107</td><td>A23CS0135</td><td>Muhammad Iqram</td><td>iqram@graduate.utm.my</td><td>Bioinformatics</td></tr><tr><td>1129</td><td>A23CS0265</td><td>Sabrina Heng</td><td>sabrina@graduate.utm.my</td><td>Data Engineering</td></tr><tr><td>1212</td><td>A23CS0029</td><td>Yusuf</td><td>yusuf@graduate.utm.my</td><td>Graphics and Multimedia Software</td></tr><tr><td>1213</td><td>A23CS9000</td><td>Transaction Test Student</td><td>test@graduate.utm.my</td><td>Software Engineering</td></tr></table>	student_id	matric_no	name	email	programme	1001	A23CS0009	Aisyah	aisyah@graduate.utm.my	Computer Networks and Security	1008	A23CS0018	Brendan Chia	brendan@graduate.utm.my	Bioinformatics	1017	A23CS0037	Fatin Alfah	fatin@graduate.utm.my	Computer Networks and Security	1023	A23CS0080	Gui Kah Sin	kahsin@graduate.utm.my	Data Engineering	1056	A23CS0098	Lau Yan Kai	yanikai@graduate.utm.my	Software Engineering	1080	A23CS0128	Matthew	matthew@graduate.utm.my	Data Engineering	1107	A23CS0135	Muhammad Iqram	iqram@graduate.utm.my	Bioinformatics	1129	A23CS0265	Sabrina Heng	sabrina@graduate.utm.my	Data Engineering	1212	A23CS0029	Yusuf	yusuf@graduate.utm.my	Graphics and Multimedia Software	1213	A23CS9000	Transaction Test Student	test@graduate.utm.my	Software Engineering
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2.4 Explanation of SQL Files

The back-end of the academic attendance monitoring system is implemented with a set of structured SQL scripts. The structure of these files is designed to provide different modules, easy execution and data integrity.

I. Database Definition (DDL)

The database definition (DDL) core structure is defined in the initialisation script. This script is responsible for:

#	Structure	Description
1.	Schema Creation	Creating the <i>attendance_system</i> database
2.	Table Construction	Defining the six core tables (<i>students</i> , <i>lecturers</i> , <i>courses</i> , <i>classes</i> , <i>enrollments</i> , and <i>attendance_records</i>)
3.	Constraint Enforcement	Applying Primary Keys (AUTO_INCREMENT), Unique Keys (for <i>matric_no</i> , <i>staff_no</i> , <i>email</i>), and Foreign Keys to maintain Referential Integrity between entities, for example linking enrolments to students and classes.

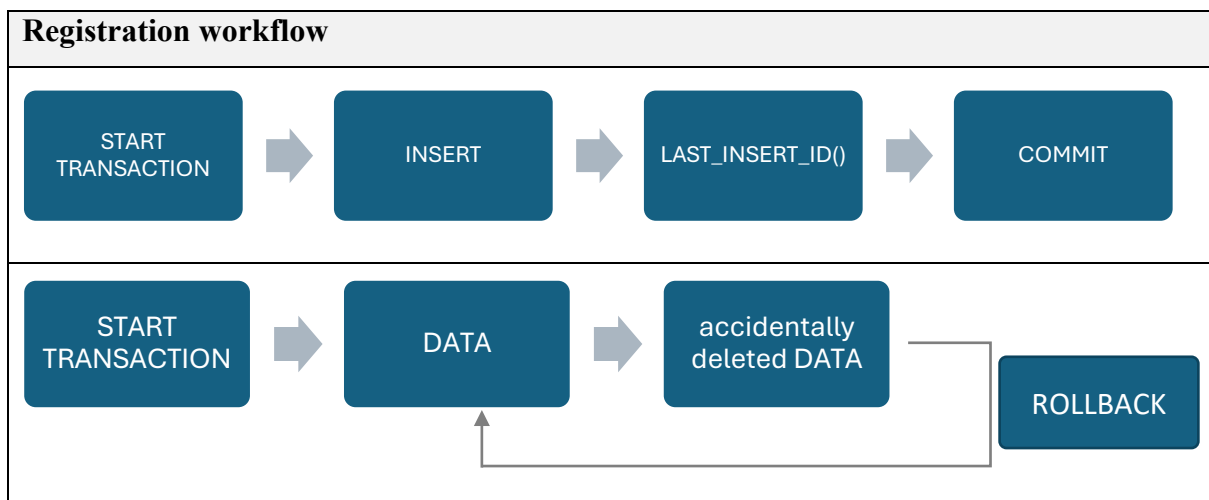
II. Data Manipulation (DML)

The data population script handles the overall data processing for the database. It includes:

#	Structure	Description
1.	Bulk Insertion	Populating the tables with sample data (10 students, 10 lecturers, 14 classes, and associated enrolments)
2.	Automated Logic	Executing UPDATE statements to automatically mark students as 'Late' if their <i>attendance_time</i> exceeds the class start time
3.	Cleanup	Implementing DELETE operations with ON DELETE CASCADE to show how the system handles student withdrawals without leaving orphaned records

III. Transaction Management

We use atomic transaction to ensure the reliability of data in the transaction process.



This ensures that students can only create classes if they successfully register for classes, thus avoiding data inconsistencies.

IV. Advanced Querying (DQL)

The last set of scripts includes the analysis queries used in the report. Aggregation for statistics of students, filtering for identifying late attendees and complex connections for linking students with their attendance history are used to extract important information from raw data.

QUERY DEMONSTRATION

3.0 Filtering and Sorting

This section will show the system ability to retrieve specific data subsets using conditional filtering and organise the results using sorting mechanisms.

3.1 Aggregation and Grouping

SQL Code	Output																				
1. Filtering																					
Use the combination of WHERE , AND , and OR to filter and retrieve students who arrived late with the condition enrolled in either 'Data Engineering' or 'Software Engineering' programme. The results then sorted by programme																					
<pre># Filtering (WHERE, OR & AND) # List late students from specific programmes SELECT s.student_id, s.name, s.programme, a.attendance_date, a.attendance_time, a.remark FROM attendance_records a JOIN enrollments e ON a.enroll_id = e.enroll_id JOIN students s ON e.student_id = s.student_id WHERE a.status = 'Late' AND (s.programme = 'Data Engineering' OR s.programme = 'Software Engineering') ORDER BY s.programme;</pre>	<table><tr><th>student_id</th><th>name</th><th>programme</th><th>attendance_date</th><th>attendance_time</th><th>remark</th></tr><tr><td>1080</td><td>Matthew</td><td>Data Engineering</td><td>2025-11-11</td><td>08:15:00</td><td>Traffic delay</td></tr></table>	student_id	name	programme	attendance_date	attendance_time	remark	1080	Matthew	Data Engineering	2025-11-11	08:15:00	Traffic delay								
student_id	name	programme	attendance_date	attendance_time	remark																
1080	Matthew	Data Engineering	2025-11-11	08:15:00	Traffic delay																
2. Sorting																					
Retrieve the first two (LIMIT 2) lectures who belong to the 'Department of Multimedia' and sorted alphabetically by their names (ORDER BY name ASC).																					
<pre># Sorting (ORDER BY, LIMIT) SELECT * FROM lecturers WHERE department = 'Department of Multimedia' ORDER BY name ASC LIMIT 2;</pre>	<table><tr><th>lecturer_id</th><th>staff_no</th><th>name</th><th>department</th><th>email</th></tr><tr><td>3006</td><td>7963</td><td>Dr. Faridah</td><td>Department of Multimedia</td><td>faridah@utm.my</td></tr><tr><td>3005</td><td>15946</td><td>Dr. Hani</td><td>Department of Multimedia</td><td>hani@utm.my</td></tr><tr><td>NULL</td><td>NULL</td><td>NULL</td><td>NULL</td><td>NULL</td></tr></table>	lecturer_id	staff_no	name	department	email	3006	7963	Dr. Faridah	Department of Multimedia	faridah@utm.my	3005	15946	Dr. Hani	Department of Multimedia	hani@utm.my	NULL	NULL	NULL	NULL	NULL
lecturer_id	staff_no	name	department	email																	
3006	7963	Dr. Faridah	Department of Multimedia	faridah@utm.my																	
3005	15946	Dr. Hani	Department of Multimedia	hani@utm.my																	
NULL	NULL	NULL	NULL	NULL																	
3. Aggregation																					

SQL Code	Output																		
Calculates the number of students in each programme using COUNT (*) and store in the total_students column. It also uses the window function OVER() to compare each programme's student count with the overall MAX and MIN across all programmes, then labeling them as 'MAX' or "MIN" in the new column named <i>student_status</i>. The result lists programme counts in descending order.																			
<pre># Aggregation (COUNT, MAX & MIN) # 'OVER()' allows MAX() and MIN() to check across all grouped rows without collapsing them into a single row SELECT programme, COUNT(*) AS total_students, CASE WHEN COUNT(*) = MAX(COUNT(*) OVER ()) THEN 'MAX' WHEN COUNT(*) = MIN(COUNT(*) OVER ()) THEN 'MIN' ELSE '' END AS student_status FROM students GROUP BY programme ORDER BY total_students DESC;</pre>	<table><tr><th>programme</th><th>total_students</th><th>student_status</th></tr><tr><td>Data Engineering</td><td>3</td><td>MAX</td></tr><tr><td>Computer Networks and Security</td><td>2</td><td></td></tr><tr><td>Bioinformatics</td><td>2</td><td></td></tr><tr><td>Software Engineering</td><td>1</td><td>MIN</td></tr><tr><td>Graphics and Multimedia Software</td><td>1</td><td>MIN</td></tr></table>	programme	total_students	student_status	Data Engineering	3	MAX	Computer Networks and Security	2		Bioinformatics	2		Software Engineering	1	MIN	Graphics and Multimedia Software	1	MIN
programme	total_students	student_status																	
Data Engineering	3	MAX																	
Computer Networks and Security	2																		
Bioinformatics	2																		
Software Engineering	1	MIN																	
Graphics and Multimedia Software	1	MIN																	
4. Grouping and filtering groups																			
Groups lecturers by name and counts how many classes each lecturer. The HAVING clause filters the result to include only lecturers who teach more than one class. The result is sorted to show the lecturers with the most classes at the top.																			
<pre># Grouping and filtering groups (GROUP BY, HAVING) # Filter those lecturer who have more than 1 class per week SELECT l.name AS lecturer, COUNT(c1.class_id) AS total_classes FROM lecturers l JOIN classes c1 ON l.lecturer_id = c1.lecturer_id GROUP BY l.name HAVING COUNT(c1.class_id) > 1 ORDER BY total_classes DESC;</pre>	<table><tr><th>lecturer</th><th>total_classes</th></tr><tr><td>Dr. Azlan</td><td>2</td></tr><tr><td>Dr. Sean</td><td>2</td></tr><tr><td>Dr. Nurin</td><td>2</td></tr><tr><td>Dr. Nies</td><td>2</td></tr></table>	lecturer	total_classes	Dr. Azlan	2	Dr. Sean	2	Dr. Nurin	2	Dr. Nies	2								
lecturer	total_classes																		
Dr. Azlan	2																		
Dr. Sean	2																		
Dr. Nurin	2																		
Dr. Nies	2																		

3.2 Functions and Conditional Logic

SQL Code	Output																																													
1. Conditional Logic																																														
<pre>SELECT course_id, class_time, CASE WHEN class_time < "10:00:00" THEN "Morning Class" WHEN "10:00:00" <= class_time <= "02:00:00" THEN "Mid-day Class" ELSE "Evening Class" END AS class_type FROM classes;</pre>	<table><tr><th>course_id</th><th>class_time</th><th>class_type</th></tr><tr><td>1</td><td>09:00:00</td><td>Morning Class</td></tr><tr><td>3</td><td>10:00:00</td><td>Mid-day Class</td></tr><tr><td>102</td><td>14:00:00</td><td>Mid-day Class</td></tr><tr><td>24</td><td>11:00:00</td><td>Mid-day Class</td></tr><tr><td>56</td><td>15:00:00</td><td>Mid-day Class</td></tr><tr><td>72</td><td>08:00:00</td><td>Morning Class</td></tr><tr><td>87</td><td>09:00:00</td><td>Morning Class</td></tr><tr><td>113</td><td>10:00:00</td><td>Mid-day Class</td></tr><tr><td>96</td><td>08:00:00</td><td>Morning Class</td></tr><tr><td>43</td><td>15:00:00</td><td>Mid-day Class</td></tr><tr><td>1</td><td>11:00:00</td><td>Mid-day Class</td></tr><tr><td>96</td><td>09:00:00</td><td>Morning Class</td></tr><tr><td>24</td><td>11:00:00</td><td>Mid-day Class</td></tr><tr><td>113</td><td>08:00:00</td><td>Morning Class</td></tr></table>	course_id	class_time	class_type	1	09:00:00	Morning Class	3	10:00:00	Mid-day Class	102	14:00:00	Mid-day Class	24	11:00:00	Mid-day Class	56	15:00:00	Mid-day Class	72	08:00:00	Morning Class	87	09:00:00	Morning Class	113	10:00:00	Mid-day Class	96	08:00:00	Morning Class	43	15:00:00	Mid-day Class	1	11:00:00	Mid-day Class	96	09:00:00	Morning Class	24	11:00:00	Mid-day Class	113	08:00:00	Morning Class
course_id	class_time	class_type																																												
1	09:00:00	Morning Class																																												
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24	11:00:00	Mid-day Class																																												
113	08:00:00	Morning Class																																												

3.3 Subqueries

SQL Code	Output						
1. Single-row subquery							
Earliest class time for the course with course_id = 1							
<pre>SELECT course_id, class_day, class_time FROM classes WHERE course_id = 1 AND class_time = (SELECT MIN(class_time) FROM classes WHERE course_id = 1);</pre>	<table><tr><th>course_id</th><th>class_day</th><th>class_time</th></tr><tr><td>1</td><td>Monday</td><td>09:00:00</td></tr></table>	course_id	class_day	class_time	1	Monday	09:00:00
course_id	class_day	class_time					
1	Monday	09:00:00					
2. Multiple-row subquery							
Lecturers who teach more than 1 class							

SQL Code	Output																														
<pre>SELECT lecturer_id, name FROM lecturers WHERE lecturer_id IN (SELECT lecturer_id FROM classes GROUP BY lecturer_id HAVING COUNT(*) > 1);</pre>	<table><tr><th>lecturer_id</th><th>name</th></tr><tr><td>3001</td><td>Dr. Azlan</td></tr><tr><td>3004</td><td>Dr. Sean</td></tr><tr><td>3008</td><td>Dr. Nurin</td></tr><tr><td>3009</td><td>Dr. Nies</td></tr></table>	lecturer_id	name	3001	Dr. Azlan	3004	Dr. Sean	3008	Dr. Nurin	3009	Dr. Nies																				
lecturer_id	name																														
3001	Dr. Azlan																														
3004	Dr. Sean																														
3008	Dr. Nurin																														
3009	Dr. Nies																														
3. Correlated subquery																															
Find the latest attendance date for each student																															
<pre>SELECT s.student_id, s.name, (SELECT MAX(a.attendance_date) FROM enrollments e JOIN attendance_records a ON e.enroll_id = a.enroll_id WHERE e.student_id = s.student_id) AS last_attendance FROM students s;</pre>	<table><tr><th>student_id</th><th>name</th><th>last_attendance</th></tr><tr><td>1001</td><td>Aisyah</td><td>2025-11-11</td></tr><tr><td>1008</td><td>Brendan Chia</td><td>2025-11-11</td></tr><tr><td>1017</td><td>Fatin Afifah</td><td>2025-11-11</td></tr><tr><td>1023</td><td>Gui Kah Sin</td><td>2025-11-11</td></tr><tr><td>1056</td><td>Lau Yan Kai</td><td>2025-11-11</td></tr><tr><td>1080</td><td>Matthew</td><td>2025-11-11</td></tr><tr><td>1107</td><td>Muhammad Iqram</td><td>2025-11-11</td></tr><tr><td>1129</td><td>Sabrina Heng</td><td>2025-11-11</td></tr><tr><td>1212</td><td>Yusuf</td><td>2025-11-11</td></tr></table>	student_id	name	last_attendance	1001	Aisyah	2025-11-11	1008	Brendan Chia	2025-11-11	1017	Fatin Afifah	2025-11-11	1023	Gui Kah Sin	2025-11-11	1056	Lau Yan Kai	2025-11-11	1080	Matthew	2025-11-11	1107	Muhammad Iqram	2025-11-11	1129	Sabrina Heng	2025-11-11	1212	Yusuf	2025-11-11
student_id	name	last_attendance																													
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1129	Sabrina Heng	2025-11-11																													
1212	Yusuf	2025-11-11																													

3.4 Set Operations

SQL Code	Output										
1. Union											
<pre>SELECT name, programme FROM students WHERE programme = 'Data Engineering' UNION SELECT name, programme FROM students WHERE programme = 'Software Engineering';</pre>	<table> <thead> <tr> <th>name</th><th>programme</th></tr> </thead> <tbody> <tr> <td>Gui Kah Sin</td><td>Data Engineering</td></tr> <tr> <td>Matthew</td><td>Data Engineering</td></tr> <tr> <td>Sabrina Heng</td><td>Data Engineering</td></tr> <tr> <td>Lau Yan Kai</td><td>Software Engineering</td></tr> </tbody> </table>	name	programme	Gui Kah Sin	Data Engineering	Matthew	Data Engineering	Sabrina Heng	Data Engineering	Lau Yan Kai	Software Engineering
name	programme										
Gui Kah Sin	Data Engineering										
Matthew	Data Engineering										
Sabrina Heng	Data Engineering										
Lau Yan Kai	Software Engineering										
2. Not Exists											
<pre>SELECT s.student_id, s.name FROM students s WHERE NOT EXISTS (SELECT 1 FROM enrollments e JOIN attendance_records a ON e.enroll_id = a.enroll_id WHERE e.student_id = s.student_id AND a.status = 'Late');</pre>	<table> <thead> <tr> <th>student_id</th><th>name</th></tr> </thead> <tbody> <tr> <td>1008</td><td>Brendan Chia</td></tr> <tr> <td>1023</td><td>Gui Kah Sin</td></tr> <tr> <td>1056</td><td>Lau Yan Kai</td></tr> <tr> <td>1129</td><td>Sabrina Heng</td></tr> </tbody> </table>	student_id	name	1008	Brendan Chia	1023	Gui Kah Sin	1056	Lau Yan Kai	1129	Sabrina Heng
student_id	name										
1008	Brendan Chia										
1023	Gui Kah Sin										
1056	Lau Yan Kai										
1129	Sabrina Heng										

3.5 Joins

SQL Code	Output																																																																	
1. Inner Join																																																																		
Finds a list of lecturers who are teaching more than one class																																																																		
<pre>SELECT classes.lecturer_id, lecturers.name FROM classes JOIN lecturers ON classes.lecturer_id = lecturers.lecturer_id GROUP BY lecturer_id HAVING COUNT(*) > 1;</pre>	<table><tr><th>lecturer_id</th><th>name</th></tr><tr><td>3001</td><td>Dr. Azlan</td></tr><tr><td>3004</td><td>Dr. Sean</td></tr><tr><td>3008</td><td>Dr. Nurin</td></tr><tr><td>3009</td><td>Dr. Nies</td></tr></table>	lecturer_id	name	3001	Dr. Azlan	3004	Dr. Sean	3008	Dr. Nurin	3009	Dr. Nies																																																							
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3004	Dr. Sean																																																																	
3008	Dr. Nurin																																																																	
3009	Dr. Nies																																																																	
2. Natural Join																																																																		
Lists all attendance records along with the student's name and programme																																																																		
<pre>SELECT student_id, name, programme, attendance_date, attendance_time FROM students NATURAL JOIN attendance_records;</pre>	<table><tr><th>student_id</th><th>name</th><th>programme</th><th>attendance_date</th><th>attendance_time</th></tr><tr><td>1212</td><td>Yusuf</td><td>Graphics and Multimedia Software</td><td>2025-11-10</td><td>09:00:00</td></tr><tr><td>1129</td><td>Sabrina Heng</td><td>Data Engineering</td><td>2025-11-10</td><td>09:00:00</td></tr><tr><td>1107</td><td>Muhammad Izam</td><td>Bioinformatics</td><td>2025-11-10</td><td>09:00:00</td></tr><tr><td>1080</td><td>Matthew</td><td>Data Engineering</td><td>2025-11-10</td><td>09:00:00</td></tr><tr><td>1056</td><td>Lau Yan Kai</td><td>Software Engineering</td><td>2025-11-10</td><td>09:00:00</td></tr><tr><td>1023</td><td>Gui Kah Sin</td><td>Data Engineering</td><td>2025-11-10</td><td>09:00:00</td></tr><tr><td>1017</td><td>Fatin Affiah</td><td>Computer Networks and Security</td><td>2025-11-10</td><td>09:00:00</td></tr><tr><td>1008</td><td>Brendan Chia</td><td>Bioinformatics</td><td>2025-11-10</td><td>09:00:00</td></tr><tr><td>1001</td><td>Aisyah</td><td>Computer Networks and Security</td><td>2025-11-10</td><td>09:00:00</td></tr><tr><td>1212</td><td>Yusuf</td><td>Graphics and Multimedia Software</td><td>2025-11-11</td><td>09:10:00</td></tr><tr><td>1129</td><td>Sabrina Heng</td><td>Data Engineering</td><td>2025-11-11</td><td>09:10:00</td></tr><tr><td>1107</td><td>Muhammad Izam</td><td>Bioinformatics</td><td>2025-11-11</td><td>09:10:00</td></tr></table>	student_id	name	programme	attendance_date	attendance_time	1212	Yusuf	Graphics and Multimedia Software	2025-11-10	09:00:00	1129	Sabrina Heng	Data Engineering	2025-11-10	09:00:00	1107	Muhammad Izam	Bioinformatics	2025-11-10	09:00:00	1080	Matthew	Data Engineering	2025-11-10	09:00:00	1056	Lau Yan Kai	Software Engineering	2025-11-10	09:00:00	1023	Gui Kah Sin	Data Engineering	2025-11-10	09:00:00	1017	Fatin Affiah	Computer Networks and Security	2025-11-10	09:00:00	1008	Brendan Chia	Bioinformatics	2025-11-10	09:00:00	1001	Aisyah	Computer Networks and Security	2025-11-10	09:00:00	1212	Yusuf	Graphics and Multimedia Software	2025-11-11	09:10:00	1129	Sabrina Heng	Data Engineering	2025-11-11	09:10:00	1107	Muhammad Izam	Bioinformatics	2025-11-11	09:10:00
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1107	Muhammad Izam	Bioinformatics	2025-11-11	09:10:00																																																														

SQL Code	Output																																																																																
3. Left Join																																																																																	
List all students along with their attendance information for the date 2025-11-10.																																																																																	
<pre>SELECT s.student_id, s.name, s.programme, a.attendance_date, a.attendance_time, a.status FROM students s LEFT JOIN attendance_records a ON a.attendance_date = '2025-11-10' ORDER BY s.student_id;</pre>	<table><tr><th>student_id</th><th>name</th><th>programme</th><th>attendance_date</th><th>attendance_time</th><th>status</th></tr><tr><td>1001</td><td>Aisyah</td><td>Computer Networks and Security</td><td>2025-11-10</td><td>11:05:00</td><td>Late</td></tr><tr><td>1001</td><td>Aisyah</td><td>Computer Networks and Security</td><td>2025-11-10</td><td>10:00:00</td><td>Present</td></tr><tr><td>1001</td><td>Aisyah</td><td>Computer Networks and Security</td><td>2025-11-10</td><td>14:00:00</td><td>Present</td></tr><tr><td>1001</td><td>Aisyah</td><td>Computer Networks and Security</td><td>2025-11-10</td><td>09:00:00</td><td>Present</td></tr><tr><td>1008</td><td>Brend...</td><td>Bioinformatics</td><td>2025-11-10</td><td>13:10:00</td><td>Late</td></tr><tr><td>1008</td><td>Brend...</td><td>Bioinformatics</td><td>2025-11-10</td><td>10:00:00</td><td>Present</td></tr><tr><td>1008</td><td>Brend...</td><td>Bioinformatics</td><td>2025-11-10</td><td>09:00:00</td><td>Present</td></tr><tr><td>1008</td><td>Brend...</td><td>Bioinformatics</td><td>2025-11-10</td><td>08:00:00</td><td>Present</td></tr></table>	student_id	name	programme	attendance_date	attendance_time	status	1001	Aisyah	Computer Networks and Security	2025-11-10	11:05:00	Late	1001	Aisyah	Computer Networks and Security	2025-11-10	10:00:00	Present	1001	Aisyah	Computer Networks and Security	2025-11-10	14:00:00	Present	1001	Aisyah	Computer Networks and Security	2025-11-10	09:00:00	Present	1008	Brend...	Bioinformatics	2025-11-10	13:10:00	Late	1008	Brend...	Bioinformatics	2025-11-10	10:00:00	Present	1008	Brend...	Bioinformatics	2025-11-10	09:00:00	Present	1008	Brend...	Bioinformatics	2025-11-10	08:00:00	Present																										
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1001	Aisyah	Computer Networks and Security	2025-11-10	11:05:00	Late																																																																												
1001	Aisyah	Computer Networks and Security	2025-11-10	10:00:00	Present																																																																												
1001	Aisyah	Computer Networks and Security	2025-11-10	14:00:00	Present																																																																												
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1008	Brend...	Bioinformatics	2025-11-10	08:00:00	Present																																																																												
4. Self-Join																																																																																	
Identify pairs of different students who were present in class on the same day																																																																																	
<pre>SELECT a1.enroll_id AS student1, a2.enroll_id AS student2, a1.attendance_date, a1.status AS status1, a2.status AS status2 FROM attendance_records a1 JOIN attendance_records a2 ON a1.attendance_date = a2.attendance_date AND a1.enroll_id < a2.enroll_id ORDER BY a1.attendance_date, a1.enroll_id;</pre>	<table><tr><th>student1</th><th>student2</th><th>attendance_date</th><th>status1</th><th>status2</th></tr><tr><td>1</td><td>3</td><td>2025-11-10</td><td>Present</td><td>Present</td></tr><tr><td>1</td><td>11</td><td>2025-11-10</td><td>Present</td><td>Absent</td></tr><tr><td>1</td><td>5</td><td>2025-11-10</td><td>Present</td><td>Present</td></tr><tr><td>1</td><td>7</td><td>2025-11-10</td><td>Present</td><td>Late</td></tr><tr><td>1</td><td>17</td><td>2025-11-10</td><td>Present</td><td>Present</td></tr><tr><td>1</td><td>9</td><td>2025-11-10</td><td>Present</td><td>Present</td></tr><tr><td>1</td><td>15</td><td>2025-11-10</td><td>Present</td><td>Present</td></tr><tr><td>1</td><td>13</td><td>2025-11-10</td><td>Present</td><td>Present</td></tr><tr><td>1</td><td>19</td><td>2025-11-10</td><td>Present</td><td>Late</td></tr><tr><td>3</td><td>5</td><td>2025-11-10</td><td>Present</td><td>Present</td></tr><tr><td>3</td><td>19</td><td>2025-11-10</td><td>Present</td><td>Late</td></tr><tr><td>3</td><td>7</td><td>2025-11-10</td><td>Present</td><td>Late</td></tr><tr><td>3</td><td>17</td><td>2025-11-10</td><td>Present</td><td>Present</td></tr><tr><td>3</td><td>9</td><td>2025-11-10</td><td>Present</td><td>Present</td></tr><tr><td>3</td><td>15</td><td>2025-11-10</td><td>Present</td><td>Present</td></tr></table>	student1	student2	attendance_date	status1	status2	1	3	2025-11-10	Present	Present	1	11	2025-11-10	Present	Absent	1	5	2025-11-10	Present	Present	1	7	2025-11-10	Present	Late	1	17	2025-11-10	Present	Present	1	9	2025-11-10	Present	Present	1	15	2025-11-10	Present	Present	1	13	2025-11-10	Present	Present	1	19	2025-11-10	Present	Late	3	5	2025-11-10	Present	Present	3	19	2025-11-10	Present	Late	3	7	2025-11-10	Present	Late	3	17	2025-11-10	Present	Present	3	9	2025-11-10	Present	Present	3	15	2025-11-10	Present	Present
student1	student2	attendance_date	status1	status2																																																																													
1	3	2025-11-10	Present	Present																																																																													
1	11	2025-11-10	Present	Absent																																																																													
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3	9	2025-11-10	Present	Present																																																																													
3	15	2025-11-10	Present	Present																																																																													

OPTIMISATION SECTION

4.0 Indexing

SQL Code	Output																																																		
1. Create BTEE index for attendance date																																																			
<pre># BTREE Index for attendance date CREATE INDEX idx_attendance_date ON attendance_records (attendance_date) USING BTREE;</pre>	<table><tr><th>Table</th><th>Nor</th><th>Key_name</th><th>Sec</th><th>Column_name</th><th>Colli</th><th>Cardis</th><th>S</th><th>P</th><th>Index_type</th></tr><tr><td>attendance_records</td><td>0</td><td>PRIMARY</td><td>1</td><td>attendance_id</td><td>A</td><td>20</td><td></td><td></td><td>BTREE</td></tr><tr><td>attendance_records</td><td>1</td><td>idx_attendance_date</td><td>1</td><td>attendance_date</td><td>A</td><td>2</td><td></td><td></td><td>BTREE</td></tr></table>	Table	Nor	Key_name	Sec	Column_name	Colli	Cardis	S	P	Index_type	attendance_records	0	PRIMARY	1	attendance_id	A	20			BTREE	attendance_records	1	idx_attendance_date	1	attendance_date	A	2			BTREE																				
Table	Nor	Key_name	Sec	Column_name	Colli	Cardis	S	P	Index_type																																										
attendance_records	0	PRIMARY	1	attendance_id	A	20			BTREE																																										
attendance_records	1	idx_attendance_date	1	attendance_date	A	2			BTREE																																										
2. Create TEXT index for name searching																																																			
<pre>CREATE FULLTEXT INDEX idx_student_name ON students (name);</pre>	<table><tr><th>Table</th><th>Nor</th><th>Key_name</th><th>Sec</th><th>Column_name</th><th>Colli</th><th>Cardis</th><th>S</th><th>P</th><th>Index_type</th></tr><tr><td>students</td><td>0</td><td>PRIMARY</td><td>1</td><td>student_id</td><td>A</td><td>9</td><td></td><td></td><td>BTREE</td></tr><tr><td>students</td><td>0</td><td>matric_no</td><td>1</td><td>matric_no</td><td>A</td><td>9</td><td></td><td></td><td>BTREE</td></tr><tr><td>students</td><td>0</td><td>email</td><td>1</td><td>email</td><td>A</td><td>9</td><td></td><td></td><td>BTREE</td></tr><tr><td>students</td><td>1</td><td>idx_student_name</td><td>1</td><td>name</td><td></td><td>9</td><td></td><td></td><td>FULLTEXT</td></tr></table>	Table	Nor	Key_name	Sec	Column_name	Colli	Cardis	S	P	Index_type	students	0	PRIMARY	1	student_id	A	9			BTREE	students	0	matric_no	1	matric_no	A	9			BTREE	students	0	email	1	email	A	9			BTREE	students	1	idx_student_name	1	name		9			FULLTEXT
Table	Nor	Key_name	Sec	Column_name	Colli	Cardis	S	P	Index_type																																										
students	0	PRIMARY	1	student_id	A	9			BTREE																																										
students	0	matric_no	1	matric_no	A	9			BTREE																																										
students	0	email	1	email	A	9			BTREE																																										
students	1	idx_student_name	1	name		9			FULLTEXT																																										

4.1 Query Analysis before Indexing

SQL Code	Output																								
1. Full table scan before BTREE index is created.																									
<pre>EXPLAIN SELECT * FROM attendance_records WHERE attendance_date = '2025-11-12';</pre>	<table><tr><th>id</th><th>select_type</th><th>table</th><th>partitions</th><th>type</th><th>possible_keys</th><th>key</th><th>key_len</th><th>ref</th><th>rows</th><th>filtered</th><th>Extra</th></tr><tr><td>1</td><td>SIMPLE</td><td>attend...</td><td>NULL</td><td>ALL</td><td>NULL</td><td>NULL</td><td>NULL</td><td>NULL</td><td>20</td><td>10.00</td><td>Using where</td></tr></table>	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra	1	SIMPLE	attend...	NULL	ALL	NULL	NULL	NULL	NULL	20	10.00	Using where
id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra														
1	SIMPLE	attend...	NULL	ALL	NULL	NULL	NULL	NULL	20	10.00	Using where														
2. Full table scan before FULLTEXT index is created.																									
<pre>EXPLAIN SELECT * FROM students WHERE name LIKE '%Brendan%';</pre>	<table><tr><th>id</th><th>select_type</th><th>table</th><th>partitions</th><th>type</th><th>possible_keys</th><th>key</th><th>key_len</th><th>ref</th><th>rows</th><th>filtered</th><th>Extra</th></tr><tr><td>1</td><td>SIMPLE</td><td>students</td><td>NULL</td><td>ALL</td><td>NULL</td><td>NULL</td><td>NULL</td><td>NULL</td><td>9</td><td>11.11</td><td>Using where</td></tr></table>	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra	1	SIMPLE	students	NULL	ALL	NULL	NULL	NULL	NULL	9	11.11	Using where
id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra														
1	SIMPLE	students	NULL	ALL	NULL	NULL	NULL	NULL	9	11.11	Using where														

4.2 Query Analysis after Indexing

SQL Code	Output																								
1. Targeted scan after BTREE index is created.																									
<pre>EXPLAIN SELECT * FROM attendance_records WHERE attendance_date = '2025-11-12';</pre>	<table><tr><th>id</th><th>select_type</th><th>table</th><th>partitions</th><th>type</th><th>possible_keys</th><th>key</th><th>key_len</th><th>ref</th><th>rows</th><th>filtered</th><th>Extra</th></tr><tr><td>1</td><td>SIMPLE</td><td>attend...</td><td>NULL</td><td>ref</td><td>idx_attendance...</td><td>id...</td><td>3</td><td>co...</td><td>1</td><td>100.00</td><td>NULL</td></tr></table>	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra	1	SIMPLE	attend...	NULL	ref	idx_attendance...	id...	3	co...	1	100.00	NULL
id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra														
1	SIMPLE	attend...	NULL	ref	idx_attendance...	id...	3	co...	1	100.00	NULL														
2. Targeted scan after FULLTEXT index is created.																									
<pre>EXPLAIN SELECT * FROM students WHERE MATCH(name) AGAINST('Brendan');</pre>	<table><tr><th>id</th><th>select_type</th><th>table</th><th>partitions</th><th>type</th><th>possible_keys</th><th>key</th><th>key_len</th><th>ref</th><th>rows</th><th>filtered</th><th>Extra</th></tr><tr><td>1</td><td>SIMPLE</td><td>students</td><td>NULL</td><td>ful...</td><td>idx_student_name</td><td>id...</td><td>0</td><td>co...</td><td>1</td><td>100.00</td><td>Using wh...</td></tr></table>	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra	1	SIMPLE	students	NULL	ful...	idx_student_name	id...	0	co...	1	100.00	Using wh...
id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra														
1	SIMPLE	students	NULL	ful...	idx_student_name	id...	0	co...	1	100.00	Using wh...														

4.3 Performance of Indexing

BTREE Indexing			
Metric	Before Indexing	After indexing	Description
Search method	ALL, full table scan	REF, targeted scan	Instead of reading the whole table, it jumped directly to the specific date
Index used	NULL	idx_attendance_date	The database used the new BTREE index we created
Rows Checked	20	1 (only matches)	Did much less work. Skipped over 19 irrelevant records
Efficiency	Slow	Fast	The query is now optimised and will stay fast

FULLTEXT Indexing			
Metric	Before Indexing	After indexing	Description
Search method	ALL, full table scan	FULLTEXT, targeted scan	Switched from full table scan to a specialised text-search engine
Index used	NULL	idx_student_name	Used the index in the back of the book approach instead of reading every name
Rows Checked	10	1 (only matches)	Found the exact student immediately without looking at the other 9 students
Efficiency	Inefficient	Highly efficient	Text searching is now scalable and won't slow down the system

Therefore, by having BTREE and FULLTEXT indexes, it successfully reduced the workload on the database. In both cases, we remove the need for Full Table Scan to make sure the academic attendance system remains responsive even the student list grows.

TEAM WORK

Team Member	Task Completed	Reflection
Brendan Chia Yan Fei [A23CS0211]	- Advanced DQL - Joins - Subqueries	From this project, I learnt how to use different types of JOINS like Inner and Left Join to connect the student and attendance tables. I also gained a better understanding of Subqueries, which helped me solve logic that required processing data in multiple steps.
Gui Kah Sin [A23CS0080]	- DML implementation - DML manipulation - Basic DQL	This project helped me understand how SQL functions produce different results depending on their usage, whether alone or joined with others. I gained hands-on experience with INSERT and UPDATE and learned the importance of using the WHERE clause carefully to avoid unintended changes. I also learned when to use HAVING and how to apply CASE statements effectively.

Team Member	Task Completed	Reflection
Sabrina Heng Wei Qi [A23CS0265]	- Database design - Database creation - DDL manipulation	Through this project, I learned how important DDL is in building a reliable database. By applying constraints like primary keys, unique fields, and NOT NULL, I realised how these rules protect the system from invalid or duplicated data. Working on the structure helped me understand that good database design is the foundation of accurate and consistent information in any real application.
Lau Yan Kai [A23CS0098]	- Transaction - Indexing - Optimisation analysis - Advanced DQL	From this project, I learned how to use Transactions to ensure that multi-step operations succeed or fail together. I also learned how to create BTREE and FULLTEXT Indexes and use the EXPLAIN command to prove that our queries are faster and more efficient.

CONCLUSION

In a word, the project used MySQL to successfully design and realise the functional database back-end of the academic attendance tracking system. We have achieved the project goals by establishing a powerful database structure (DDL), using transactions (DML) to effectively manage data, and developing advanced SQL queries (DQL) to support system operations. The implementation of BTREE and FULLTEXT indexes has further optimised the system and ensured faster data retrieval for reports and searches.

6.1 Challenge Faced

During the development process, the team faced some technical challenge. Complex queries such as those that use related subqueries and connections to generate student statistics require extensive testing to ensure accuracy. We also worked hard to understand how to use transactions to make sure that a student is registered and enrolled at the exact same time. We must also carefully check the before and after query execution plan to understand how our index can improve performance.

6.2 Suggestion for Improvement

In order to improve the system in the future, we recommend using stored procedures. This will enable us to keep common complex searches for easy reuse without reprogramming. We also recommend using database triggers to automate operations. For example, if the course is later than the scheduled time, the student's status will be automatically upgraded to "late". Also, we suggest creating a user-friendly frontend interface will enable lecturers and administrators to use this backend technology in the real world.

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